

PAPER_1333 — Dark Age 21cm EDGES Signal

Author: Daniel T. Murphy **Framework:** UQFF — Star-Magic v5.27+ **Tier:** D — Astrophysics / Cosmology Open (CLOSED) **Date:** June 11, 2026 **Calculator:** `calculate_paradox({"paradox": "dark_age_21cm"})`

Master Expression

$$T_{21\text{cm}} = -D_{\text{phys}} \times A_5 \times \beta_i \times 2 = -289 \text{ mK}$$

Match: between standard -200 mK and EDGES -500 mK

Interpretation

Deeper 21cm absorption signal via DPM buoyancy interaction at $z=17$.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

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